siox

A hardware description language for the whole circuit

Purpose and design · October 2026

Abstract. siox ("silicon oxide") is a hardware description language and compiler. It describes digital circuits with the precise semantics of VHDL and the structure of Rust: entities and processes, nine-value IEEE logic and delta cycles, written with modules, traits, generics and one strict type system. Designs compile through LLVM to native simulators that run their own tests and write VCD or FST waveforms.

This document explains what siox is for and what it will do. Part I gives the purpose and the principles. Part II describes the digital language as it exists today. Part III describes where it is going: the language features already designed, analogue and mixed-signal modelling, and the design and synthesis layer that turns models into hardware. Each feature is marked with where it stands: implemented, partly implemented, accepted, not implemented, proposal, on hold, Phase 2 or Phase 3.

Contents

Part I · Purpose 5
1 Why another HDL 6
1.1 What “the whole circuit” means 6
1.2 Who siox is for 6
2 Principles 7
2.1 Hardware semantics first 7
2.2 Two audiences, one language 7
2.3 Three sigils, one job each 7
2.4 The library over the compiler 7
2.5 Tooling that behaves like a modern compiler 8
2.6 Deterministic 8
2.7 Phased, and honest about it 8
3 A first look 9
Part II · The digital language 11
4 Program structure 12
4.1 Modules and imports 12
4.2 Entities and implementations 12
4.3 Instances and connections 13
4.4 Instance arrays 13
4.5 Buses and views 14
4.6 Parameters and generation 14
5 Behaviour 16
5.1 Wires and processes 16
5.2 One assignment operator 16
5.3 Events, clocks and reset 17
5.4 Conditions 17
5.5 Pattern matching 17
5.6 Multiple drivers and resolution 18
5.7 Time 18
6 Types 20
6.1 The kernel 20
6.2 Logic values 20
6.3 Numeric vectors 20
6.4 Fixed and floating point 21
6.5 Ranged integers 21
6.6 Aggregates 21
6.7 Arrays 21
6.8 Newtypes and aliases 22
6.9 Every value starts defined 22
7 Abstraction 23
7.1 Traits and generic implementations 23
7.2 Operators 23
7.3 Literal prefixes and suffixes 24
7.4 Conversions: `From`, `Into` and `Default` 24
7.5 `#[derive]` 25
7.6 Type attributes 25
7.7 Functions 25
8 Metadata and directives 27
8.1 Declared attributes 27
8.2 Directives 27

8.3 Lints ... 28
9 Verification ... 29
9.1 Testbenches are entities ... 29
9.2 The test executable ... 30
9.3 Checking and reporting ... 30
9.4 Stimulus services ... 31
9.5 Waveforms and debugging ... 31
10 The compiler ... 32
10.1 One pipeline ... 32
10.2 Native simulation ... 32
10.3 Diagnostics ... 32
10.4 C functions ... 33
10.5 Embedding and editors ... 33
Part III · Where siox is going ... 34
11 The digital language, growing ... 35
11.1 Pipelined functions ... 35
11.2 Macros ... 36
11.3 `core` and `std` ... 37
11.4 The standard library ... 37
11.5 Entity methods ... 38
11.6 Implicit receivers and `static` ... 38
11.7 Scoped blocks ... 39
11.8 Payload enums ... 40
11.9 Binary and unary custom operators ... 40
11.10 The last implicit conversion site ... 41
11.11 Compiler foundations ... 41
11.12 Parallel simulation ... 41
11.13 Co-simulation and the simulator interfaces ... 42
12 Phase 2: analogue and mixed signal ... 44
12.1 What it will do ... 44
12.2 What it will not do ... 44
12.3 How it will be judged ... 44
13 Phase 3: design and synthesis ... 45
13.1 Projects and libraries ... 45
13.2 Foreign HDL, side by side ... 45
13.3 Constraints and physical intent ... 45
13.4 Synthesis through a vendor-neutral RTL ... 45
13.5 Schematics and boards ... 46
13.6 How it will be judged ... 46
14 Tools, boundaries and status ... 47
14.1 The tools around the compiler ... 47
14.2 What siox will not be ... 47
14.3 How siox compares ... 47
14.4 Where things stand ... 47
14.5 Summary ... 48
Appendix ... 49
15 Reference ... 50
15.1 Keywords ... 50
15.2 System attributes ... 50
15.3 Directives ... 50
15.4 Lints ... 51
15.5 Glossary ... 51

15.6 Further reading 51

Part I · Purpose

1 Why another HDL

Hardware engineers today choose between languages that each solve part of the problem.

- **VHDL** has the most careful semantics of the mainstream HDLs. Its nine-value logic, resolved signals and delta-cycle model describe real circuits precisely. Its syntax and type system date from the 1980s, and writing reusable, well-structured code in it is laborious: generics are limited, there are no traits or modules in the modern sense, and the tooling around it is old.
- **Verilog and SystemVerilog** are concise and universally supported, but their semantics are loose: four-value logic, implicit widening and truncation, scheduling races between blocks, and a well-known gap between what simulates and what synthesizes. SystemVerilog added a large verification language on top rather than tightening the design language underneath.
- **Embedded generators** such as Chisel and Amaranth use a general-purpose language to build circuits. They gain that language's abstractions, but the circuit is a data structure the host program constructs, so errors, types and tooling belong to the host, not to the hardware, and simulation is a separate world from generation.
- **Newer HDLs** such as Spade and Clash bring modern language design to hardware. siox shares their direction and borrows from them where they have already solved a problem; Spade's pipelines (§ 11.1) are one example.

None of them covers more than the digital part of a circuit. A real board mixes digital logic with analogue parts, power, clocks and the physical connections between them, and today each of those is described in a different tool, in a different language, by a different team.

> *siox aims to be one language for the whole circuit: precise enough to simulate digital logic exactly, extensible to analogue behaviour, and structured enough to compose both into a design that existing tools can build.*

1.1 What "the whole circuit" means

The project is organised around one rule:

> *Digital and analogue define components. Design connects and realizes them.*

- **Components** are described in the language: a counter, a UART, a PLL model, an amplifier stage, a power rail. Digital components exist today; analogue components are the second phase.
- **Design** is the act of connecting components into a product and handing it to the tools that build it: synthesis for FPGAs and ASICs, netlists and schematics for boards, constraints and pin assignments. That is the third phase.

Each phase builds on the one before without changing what it means. The digital language does not become approximate when analogue arrives, and the design layer does not reinterpret what the components say.

1.2 Who siox is for

- **Hardware engineers** who know VHDL or Verilog and want a language with modern structure, strict types and good tools, without giving up the hardware model they already think in.
- **Software engineers** moving into hardware, who know Rust's style and need a language whose rules catch the mistakes hardware punishes: width mismatches, multiple drivers, latches, uninitialised values.
- **Teams** that build boards and systems, not just chips, and today keep the digital design, the analogue models and the board in separate worlds.

2 Principles

2.1 Hardware semantics first

siox is strict where hardware is strict.

- Every declaration states its type. `let x = e;` without a type is an error.
- Assignment and connection widths must match exactly, and arithmetic does not widen silently: `unsigned[8] + unsigned[8]` wraps at eight bits. Widening and narrowing are written as conversions. Two numbers of different types meet in a format that holds both exactly, and a conversion happens implicitly only where a library declared it with `Into`, never to make an operator fit.
- A signal has one driver unless its type says how parallel drivers combine. Two drivers of an unresolved signal are an error that points at both.
- The logic type is the full IEEE 1076-2019 nine-value `std_ulogic` set (`U X 0 1 Z W L H -`), and uninitialized and conflicting values propagate through vectors and arithmetic as the standard defines.
- Every signal starts at a defined value: its type's default. A `Logic` starts at `'U'`, so an undriven signal reads as uninitialized rather than as a plausible `'0'`.

Behaviour is checked against an independent reference simulator (nvc for VHDL semantics), not against siox's own output.

2.2 Two audiences, one language

siox is written to be familiar to two groups at once.

- From **Rust** it takes modules and `use` imports, `pub` visibility, structs and enums, traits with generic implementations, `impl` blocks, expression syntax including `if` and `match` as expressions, newtypes, and a compiler whose diagnostics and lint levels work like rustc's.
- From **VHDL** it takes the hardware model: entities with directional ports, concurrent processes with sequential bodies, resolved signals, labels on processes and generate blocks, and attributes read with the tick (`clk'event`, `x'length`).

Where the two disagree, siox follows Rust for program structure and VHDL for circuit meaning. When siox's intended behaviour is unclear, the first question is what rustc does, and the answer stops where hardware starts.

2.3 Three sigils, one job each

The punctuation of the language has fixed meanings:

Sigil	Meaning	Example
`.`	values: fields and method calls	`pkt.data`, `clk.rising()`
`::`	types and modules: paths, variants, associated items	`std::logic::Bit`, `State::Idle`
`'`	attributes: properties of a declaration or signal	`sig'event`, `x'old`, `probe'keep`

A reader can tell from the punctuation alone whether an expression reads a value, names a type, or asks a question about a declaration.

2.4 The library over the compiler

Behaviour belongs in source code, not in the compiler. The logic values, their truth tables, numeric vectors and their operators, time units, encodings and conversions are declared in the standard library as ordinary types and trait implementations. The compiler provides the mechanisms (type checking, operator dispatch, elaboration, code generation) and evaluates the library's definitions; it does not carry a second copy of what `and` means for `Logic`. A user can define new value types, operators and literal suffixes the same way the standard library does.

The rule cuts the other way too: what only the compiler can do belongs to the compiler, and is reached through the language in a small, fixed set of places: directives, macros, and the hooks the standard library implements (§ 11.3).

2.5 Tooling that behaves like a modern compiler

`sioxc` is one compiler invocation per file, like `rustc`. It never runs what it builds. A design compiles to a native object; a file with `#[test]` entities compiles to a native test executable that runs its testbenches, filters them by name and writes waveforms. Every diagnostic carries a stable code, warnings are lints with levels set by `#[allow]`/`#[deny]` or on the command line, and the same compiler library powers an editor language server.

2.6 Deterministic

The same source always produces the same IR, the same object and the same simulation, on every machine and every run. Simulation order inside a delta cycle never depends on hash order or thread timing, randomization is seeded and reproducible, and the corpus of example programs is checked for byte-identical output between runs.

2.7 Phased, and honest about it

siox grows in phases and never accepts syntax it cannot yet give meaning to. Analogue constructs are reported as errors in the digital compiler, not ignored. Features that are designed but not built live as written proposals until they land, and removed syntax is an error that names its replacement (`process name { … }` became `name: process { … }`, `using` became `use` and `type`).

3 A first look

An eight-bit counter with a testbench:

```
module counter;

entity Counter {
    clk: Bit in,
    rst: Logic in,
    count: unsigned[8] out,
}

impl Counter {
    let value: unsigned[8] = 0;

    update: process {
        if clk.rising() {            // runs only on a rising clock edge
            if rst == '1' { value = 0; }
            else { value = value + 1; }
        }
    }

    count = value;                   // a wire: always equal to `value`
}

#[test]
entity CounterTest {}

impl CounterTest {
    let clk: Bit = '0';
    let rst: Logic = '1';
    let count: unsigned[8];
    let dut: Counter = { .clk = clk, .rst = rst, .count = count };

    clock: process {
        clk = not clk after 5ns;     // free-running clock, 10 ns period
    }

    stimulus: process {
        await 10ns;                  // hold reset for one edge
        rst = '0';
        for i in 0..9 { await clk.rising(); }
        assert!(count == 10, "counter should reach 10");
    }
}
```

The `entity` declares the interface: three ports, each a name, a type and a direction. The `impl` gives the behaviour. Two kinds of logic sit side by side: a concurrent assignment (`count = value;`) is a wire that always equals its expression, and a process is ordered behaviour, here clocked. `update:` is an optional VHDL-style label that names the process in diagnostics and tools.

The testbench is an entity too, marked with the `#[test]` directive. It instantiates the counter the way a struct literal is written, drives a clock with a delayed self-assignment, and steps through time with `await`.

```
$ sioxc --test counter.siox -o counter-tests
$ ./counter-tests -o counter.vcd
```

```
running 1 test
test counter::CounterTest ... ok

test result: ok. 1 passed; 0 failed; 0 filtered out
```

`sioxc` compiles; the executable it produces runs the tests and writes the waveform. The same separation as `rustc --test`: the compiler never runs what it builds.

Figure 1 shows what `counter.vcd` holds.

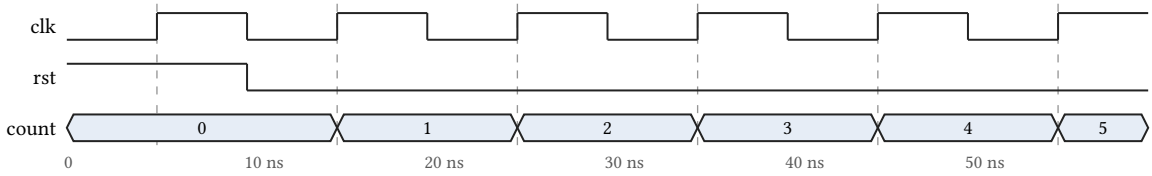


Figure 1: The first 60 ns of `CounterTest`, drawn from the VCD the test executable writes. The clock rises at 5 ns and every 10 ns after; the edge at 5 ns still sees reset, so the count starts at the edge at 15 ns.

Part II · The digital language

4 Program structure

4.1 Modules and imports

implemented

A file is one module, named on its first line. Imports follow Rust, with one change: an import is renamed with `=`, not `as`.

```
module bus::uart;

use std::logic::Logic;                  // one item
use std::bits::{unsigned, sext};        // several from one module
use Master = bus::axi::Master;          // renamed import (Rust: `as Master`)
use std::{math::{self, PI}, numeric::{Word8 = Byte}}; // nested groups
use bus::axi::*;                        // glob: every public name
use self::State::{Idle, Busy};          // enum variants, from this module
use super::spi::Clock;                  // from the parent module
pub use std::logic::{Bit, Logic};       // re-export

type Word = unsigned[32];               // transparent type alias
pub type Byte = integer<0..255>;
type Pair<T> = Packet<T>;               // generic alias
```

A `use` of a project module loads the file it names (`use bus::spi::Master;` reads `bus/spi.siox`), and `std::` paths come from the standard library. Fully qualified paths work everywhere without an import. A small prelude is in scope in every module, so `Bit`, `Logic`, `unsigned`, `signed`, `string`, `time` and the directives need no import.

Items are private to their module by default and exported with `pub`. Visibility follows the owning container, as in Rust: a struct field or an inherent method is private unless marked `pub`, and nothing can be more visible than its owner.

The rest of Rust's import model works the same way. Importing a module binds its name as a prefix (`use std::math;`, then `math::max(a, b)`). A glob is weaker than any explicit name, and two globs that bring the same name are an error only where the name is used. A `use` may open any block, and its names shadow the module's until the block ends. Importing a variant does not import its enum, and an import cannot pass through a type alias. One import binds a name in every namespace it lives in, so the module `std::assert` and the macro `assert!` never collide.

4.2 Entities and implementations

implemented

An entity is a hardware component's interface: its ports, and nothing else. Its behaviour lives in an implementation, as a Rust type's methods live in an `impl`.

```
entity Uart<BAUD: integer> {
    clk: Bit in,
    rx: Bit in,
    tx: Bit out,
    data: unsigned[8] out,
    valid: Bit out,
}

impl<BAUD: integer> Uart<BAUD> {
    let shift: unsigned[8] = 0;
```

```
    let count: integer<0..15> = 0;
    // processes, concurrent assignments, instances, functions ...
}
```

A port is written like a struct field, `name: Type`, followed by its direction: `in`, `out` or `inout`. A direction is a permission: an implementation may write its `out` ports and only read its `in` ports. Writing an input is an error. `inout` ports are tristate nets whose parallel drivers resolve.

State, behaviour and sub-instances all live in the implementation. An entity body that holds anything but ports is an error: the interface stays readable at a glance, and an implementation can be split across several `impl` blocks.

4.3 Instances and connections

implemented

An instance is declared with `let`, and its ports are connected the way a struct literal is written: by name, or by position.

```
let rx_uart: Uart<BAUD = 115200> = { .clk = clk, .rx = pin, .tx = tx, .data =
d, .valid = v };
let tx_uart: Uart<BAUD = 115200> = { clk, rx2, tx2, d2, v2 };   // by port order
```

Ports can also be wired after the declaration, through the instance:

```
let dut: Counter<W = 8>;
dut.clk = clk;          // drive an input
count = dut.count;      // read an output
```

A connection may be any value of the port's type: a signal, a slice of an array (`.inp = taps[1..2]`, whose positions connect in order), or a literal, however deeply nested (`.m = [[1, 2], [3, 4]]`, `.b = Bundle { .arr = [1, 2] }`).

A missing required connection, or a connection to a port the entity does not have, is an error naming the port; an output connected to two producers is the conflicting-driver error, pointing at both connection sites.

4.4 Instance arrays

implemented

`let stage: Cell[4];` declares four instances, `stage[0]` to `stage[3]`, and builds every one of them. Arrays nest outer index first (`Cell[2][3]` is three rows of two, read `grid[row][col]`). Writing a slot is connection sugar, not construction:

```
impl Chain {
    let stage: Cell[4];
    for i in 0..3 {
        stage[i] = { .din = taps[i] };    // connects slot i's ports
        outs[i] = stage[i].dout;          // reads a slot's output
    }
}
```

`stage[i] = Cell { .din = x }`, the name-less `stage[i] = { .din = x }`, and port writes through the slot (`stage[i].din = x`) all connect; `stage[i] = {}` connects nothing. A slot nothing wires still exists and simulates on its default inputs, which shows as a flat line in the waveform. That is the warning `unwired_instance` when the array is wired element by element, as VHDL's `open` ports are allowed but

visible. A loop over constant bounds that touches an instance array is a generate loop, in a testbench as in a design, and a runtime index selects a slot through the same multiplexing as any array.

4.5 Buses and views

implemented

A struct is layout without direction. A **view** gives a struct a protocol role, with a direction for every field, so one struct serves both ends of a bus:

```
struct Stream<T> { valid: Bit, ready: Bit, data: T }

view Source<T> for Stream<T> { valid out, ready in, data out }
view Sink<T>   for Stream<T> { valid in,  ready out, data in }

entity Producer { bus: Stream<unsigned[32]> Source }
entity Consumer { bus: Stream<unsigned[32]> Sink }

impl Top {
    let wire: Stream<unsigned[32]>;
    let p: Producer = { .bus = wire };
    let c: Consumer = { .bus = wire };
}
```

The applied type writes the backing struct first and the view second, where a port's direction goes. Views overload by backing struct, so `Source` can be declared for many protocols without a name clash, and methods and trait implementations can be attached to an applied view (`impl Stream<T> Source`).

4.6 Parameters and generation

implemented

Entities, structs, traits and functions take generic parameters, resolved at elaboration: `entity Counter<W: integer>`, instantiated as `Counter<W = 8>`. Parameters can size ports and state, select types, and drive generation.

A `for` loop or an `if` with a constant condition at the top level of an implementation is **structure**: it is unrolled or selected once, at elaboration.

```
impl Chain<N: integer> {
    let w: unsigned[8][0..N];

    stages: for k in 0..(N - 1) {
        let s: Stage = { .i = w[k], .o = w[k + 1] };
    }

    tap: if DEBUG == 1 {
        let probe: Probe = { .i = w[N] };
    } else {
        let probe: Stub = { .i = w[N] };
    }
}
```

Ranges are inclusive and directional, like a declared index range: `0..2` is 0, 1, 2 and `2..0` counts down. A labelled `for` becomes a scope in the elaborated hierarchy, with one child per iteration keyed by the loop value in square brackets (`chain.stages[0].s`); a labelled `if` is one scope filled by whichever branch is taken

(`chain.tap.probe`). The same path appears in the instance tree, the IR, the waveform and the debugger. Labels are optional; an unlabelled loop names its instances with their indices (`s_0`, `s_1`).

An entity may only be instantiated where structure is built: at the top of an implementation or inside a structural `for`/`if`. Instantiating inside a process, a `match` arm or a function is an error, because control flow that runs while simulating cannot create hardware.

A loop whose bounds are not constants cannot be unrolled into structure. Inside a process it still simulates, as a loop, with the warning `non_constant_loop_bound`: siox never refuses to simulate something because it has no hardware form, but it says so.

5 Behaviour

5.1 Wires and processes

implemented

An implementation describes behaviour two ways.

- A **concurrent assignment** at the top level is a wire: `y = a and b;` means `y` always equals `a and b`. Each one is its own driver.
- A **process** is a block of ordered statements. Processes run concurrently with each other and with every wire; the statements inside one process run in source order.

```
impl Register {
    update: process {
        if clk.rising() {
            q = d;
        }
    }

    changed = q != q'old;          // a wire beside the process
}
```

The compiler infers what a process is sensitive to; there is no VHDL-style sensitivity list. A process may carry a VHDL-style label (`update:`), and so may any assignment (`sum: y = a + b;`). Labels are never required and never change behaviour: they name things in diagnostics, the IR and the debugger.

5.2 One assignment operator

implemented

There is one assignment operator, `=`, and its meaning comes from context: an initializer on a `let`, a wire at the top level, an update inside a process, or a port connection.

Inside an event-controlled block, assignments to signals take effect at the end of the step (next-state semantics), so `a = b; b = a;` on a clock edge swaps the two values. Local `let` bindings inside a process update immediately.

Within one process, a later assignment overrides an earlier one under its condition, which allows the default-then-override style:

```
process {
    y = b;                 // default
    if sel { y = a; }      // override
}

z = if x > 200 { 200 } else if x < 10 { 10 } else { x };   // the same select as an
expression
```

There is no `?:` operator; `if` is an expression, as in Rust. Source order never gives priority **between** processes or wires: two independent drivers of one signal are either folded by the signal type's resolution, or an error.

The left side of an assignment can be part of a signal. A slice (`y[3..0] = a;`) or one element, even at a runtime position (`status[i] = err;`), is a read-modify-write that keeps the other bits; a concatenation splits one value across several places (`{carry, sum} = a + b;`); a tuple destructures (`(q, r) = divmod(x, 7);`, which also swaps `(a, b) = (b, a);` in a process). A runtime index outside the declared range is a simulation failure that reports the value, the range and the source location; it never quietly reads zero or writes nothing.

5.3 Events, clocks and reset

implemented

Every digital signal has two system attributes: `'event` (it changed in this delta cycle) and `'old` (its value before the change). Clock edges are built from them, in the standard library, as ordinary trait methods:

```
pub trait ClockLike {
    fn rising() -> Bool;
    fn falling() -> Bool;
    fn edge() -> Bool;
}
```

`clk.rising()` is defined in the standard library from `clk'event` and `clk'old`, the way VHDL's `rising_edge` is. There is no special clock type: any single-bit signal is a clock when an edge is taken on it, and a user type becomes a clock by implementing `ClockLike`.

Reset is ordinary logic. A synchronous reset is a test inside the clocked branch; an asynchronous reset is a test before it:

```
process {
    if rst == '1' { q = 0; }
    else if clk.rising() { q = d; }
}
```

5.4 Conditions

implemented

A condition must be a type that says how to be true: it implements the `Boolean` trait. `Bool` and `Bit` do. `Logic` deliberately does not, because a `Logic` can be `'X'` or `'Z'`: write `rst == '1'`, never `if rst`. A user type (a state enum, say) becomes usable as a condition by implementing `Boolean`.

5.5 Pattern matching

implemented

`match` works on enums, characters, numbers, ranges, bit patterns and tuples, with or-patterns and a wildcard, as a statement or as an expression. The first arm that matches wins, as in an `if` chain.

```
match state {
    State::Idle => { if start { state = State::Busy; } }
    State::Busy | State::Wait => { if done { state = State::Idle; } }
}

match opcode {
    "00------" => op = Op::Alu,      // '-' matches any element
    x"A?"      => op = Op::Ext,      // '?' masks a whole nibble
    _          => op = Op::Nop,
}

match (state, start, count) {        // several values at once
    (State::Idle, '1', _)   => next = State::Run,
    (State::Run,  _, 0..14) => next = State::Run,
    _                       => next = State::Idle,
}
```

```
match n {
    ..-1         => sign = Sign::Negative,   // an open range
    0..DEPTH - 1 => slot = n,
    DEPTH..      => overflow = true,
}
```

- **Bit patterns** match element by element through the element type's `Match` trait, VHDL-2008's matching equality: for `Logic` that is `std_match`, so `'H'` reads as `'1'` and a metavalue matches nothing but `-`.
- **Ranges** match through the type's `Ord`, so `time` (`0ns..9ns`) and the fixed-point formats work too; either end may be open, and a range is a set (`9..3` is `3..9`).
- **Tuples** match when every position does, so they combine all of the above.
- **Coverage** is checked: a match that misses values is `non_exhaustive_match`, an arm the earlier ones already cover is `unreachable_match_arm`, and two ranges sharing exactly one endpoint (`0..10` then `10..20`) is `overlapping_range_endpoints`, usually an off-by-one. A match that names every enum variant, every combination of a tuple of enums, or every value of a number needs no `_`. The same proof tells the latch lint that a signal every arm assigns is not a latch.

5.6 Multiple drivers and resolution

implemented

A signal driven from two places (two processes, two wires, two instance outputs) is only legal if its type implements `Resolve`, a commutative fold of the drivers' values. `Logic` implements the IEEE `resolved` function, so a tristate bus of `Logic` works; `Bit` and `ULogic` do not, so two drivers of either are an error that names each source.

```
impl Shared {
    line = a;     // two drivers of one Logic line,
    line = b;     // folded by Logic's Resolve implementation
}
```

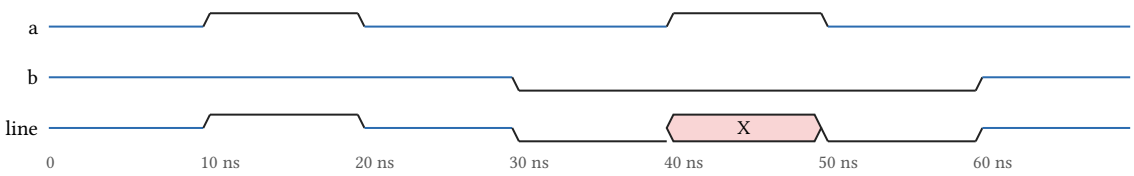


Figure 2: Two drivers of one `Logic` line, from the simulator's VCD. A released driver (`'Z'`) yields to the other; when both drive, `'1'` against `'0'` resolves to the unknown `'X'`, and the line floats at `'Z'` when neither does.

5.7 Time

implemented

Simulation time is a value: `time` is a nominal integer of femtoseconds, with unit suffixes (`10ns`, `2us`), and `frequency` a nominal real (`100MHz`).

- `await 10ns;` advances time, `await clk.rising();` waits for an edge, and `await cond;` waits until a condition holds; `await cond for 8us;` waits at most 8 µs, as VHDL's `wait until … for …`. `await` is for testbenches.
- `while cond { … }` repeats while a condition holds. Its length is decided at run time, so like `await` it is for testbenches and the functions they call; hardware loops are `for` loops with constant bounds.
- `x = v after 5ns;` schedules a delayed write with VHDL's inertial semantics. It is testbench stimulus, not hardware; in a design it is an error.

The scheduler runs delta cycles (Figure 3). Within one simulation time, wires are first settled to a fixed point; then every clocked block whose trigger changed runs, reading the values from before the edge; then all their writes are committed together. If a commit changed anything, another delta cycle follows; when nothing changes, time advances to the next scheduled event.

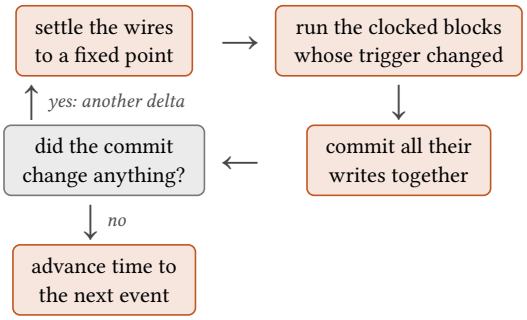


Figure 3: One delta cycle, repeated until a commit changes nothing; only then does time advance.

Because wires settle inside each delta, a wire never lags the register it reads. Figure 4 stretches the instant of one clock edge of the counter in Figure 1: the edge is delta 0, and the register `value` and the wire `count = value` both change in delta 1. (A VHDL concurrent assignment would follow one delta later.)

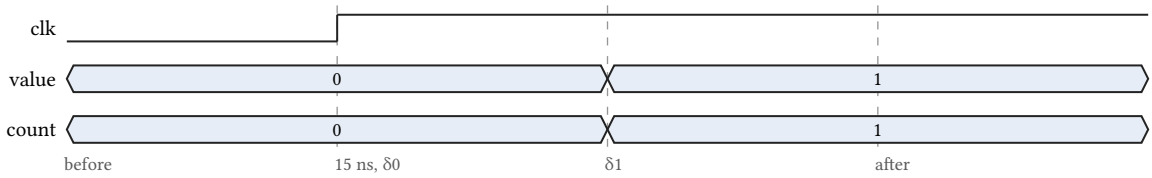


Figure 4: The clock edge at 15 ns in `Counter`, one column per delta cycle. The simulator confirms it: a process woken by `value` changing already sees the new `count`.

6 Types

6.1 The kernel

implemented

The compiler knows only a handful of types:

- `integer`, a signed 64-bit **mathematical** number for arithmetic and parameters. Leaving its range is an error, never a wrap: a division by zero or a `real` too large to convert is reported at the operation, as in VHDL.
- `natural`, its unsigned counterpart (0 to 2^{64} - 1), with the same errors (`n - 1` at zero). It mixes with `integer` only through `natural(i)` and `integer(n)`, which check the range.
- `real`, IEEE binary64: `1.0 / 0.0` is infinity and `0.0 / 0.0` is NaN.
- `Char`, a Unicode code point, deliberately not a number: a character has a number only relative to an encoding table (`Unicode::code(c)`, `Ascii::char(65)`).
- `Bool` (`true`/`false`, the type of conditions) and `string`, an array of `Char`.

Everything else, including the logic values and the numeric vectors hardware is written in, is declared in the standard library. Only `unsigned` and `signed` wrap: they are hardware words, and their arithmetic is modular.

6.2 Logic values

implemented

The digital scalars form a chain of newtypes, each adding values or behaviour:

Type	Values	Meaning
`Bit`	`'0' '1'`	Two-state, like VHDL `bit`. Internal signals and clocks.
`ULogic`	`'U' 'X' '0' '1' 'Z' 'W' 'L' 'H' '-'`	IEEE `std_ulogic`: nine values, unresolved.
`Logic`	as `ULogic`	IEEE `std_logic`: the same values, plus resolution, so tristate buses work.

The truth tables for `and`, `or`, `not`, `xor`, `nand`, `nor` and `xnor`, and the resolution function, are the IEEE 1076-2019 tables, written in the standard library as operator implementations and checked row for row against a reference simulator. How a logic value is encoded (its value bit, which values are unknown, X01 normalization) is a standard-library contract, `LogicEncoding`, evaluated by the compiler rather than duplicated in it.

A vector of logic carries its metavalues: the simulator keeps, beside each value plane, a companion plane for the elements that are not plain `'0'`/`'1'`, only where such values can actually occur. `'U'` and `'X'` propagate through logic, arithmetic and comparisons as the standard says, and waveforms show them as `x` and `z`.

6.3 Numeric vectors

implemented

`unsigned[N]` and `signed[N]` are vectors of `Logic` with a numeric meaning, like `ieee.numeric_std`. They are library types: nothing in the compiler tracks signedness, and `signed` differs from `unsigned` only by its operator implementations (a sign-aware `Ord`, an arithmetic `>>`, a signed `/`).

- **Widths are strict.** An assignment or connection must match widths exactly. Arithmetic on one type does not widen: `unsigned[8] + unsigned[8]` is `unsigned[8]` and wraps.
- **Mixed numbers promote exactly.** Two numbers of different types meet at the higher rung of one ladder, `integer` → `unsigned`/`signed` → `ufixed`/`sfixed` → `float` → `real`, in a format that holds both:

`unsigned[8] + signed[8]` is a `signed[9]`, `sfixed<8, 4> + unsigned[8]` an `sfixed<13, 4>`. Below `float` a promotion is only wiring. Storing the result still takes a conversion, where rounding and overflow are chosen.

- **Conversions.** `unsigned[16](a)` zero-extends or truncates, `signed[16](s)` extends the sign, and `resize(x, n)` takes the width as a value. A constant that does not fit its target is a compile error.
- **Arbitrary width.** Vectors are not limited to a machine word: storage, shifts (`unsigned[128](1) << 64`), metavalues and printing keep every bit of values wider than 64.
- **Literals.** `x"AB_CD"` (hex), `o"17"` (octal) and `"1X10"` (one logic value per character) are bit strings; their width is their digit count.
- **Bit methods**, after Rust's integer methods, work at any width in hardware and testbenches: `count_ones`, `leading_zeros`, `trailing_ones`, `reverse_bits`, `pow`, `saturating_add`, `rotate_left`, `is_power_of_two` and their relatives.

6.4 Fixed and floating point

implemented

The formats are type parameters. `ufixed<8, 4>` and `sfixed<8, 4>` have eight bits, four of them fraction, after VHDL's `fixed_pkg`; `float<32, 23>` is IEEE binary32 (32 bits, 23 of them mantissa), after `float_pkg`. Arithmetic, comparison and division keep the format; the format's own constructor converts a number or another format, rounding and saturating (`ufixed<8, 4>(2.5)`, `ufixed<12, 6>(x)`, `float<32, 23>(1.5)`). `float` adds square root and conversions to and from the fixed formats. All of it is library source and runs in hardware and testbenches alike.

6.5 Ranged integers

implemented

`integer<lo..hi>` is an integer restricted to a range, stored in the fewest bits that hold it (two's complement when the range goes below zero). The standard library names the familiar ones: `Byte = integer<0..255>`, `Short`, `Int`, `Long` and `Positive`. A constant out of range is a compile error; a value that leaves its range while simulating is reported with the signal's path and the declared domain.

6.6 Aggregates

implemented

- **Structs** are named fields, with positional, named and spread-update literals: `{ ..base, .valid = '1' }`.
- **Enums** are finite states. Their variants can be names (`Idle`, `Busy`) or characters, which is how `Bit` and `Logic` themselves are declared.
- **Tuples** group values by position: `(unsigned[8], Bit)` is a type, `(q, '1')` a value, `t.0` a position, and `let (q: A, r: B) = divmod(x, 7);` declares both names. A tuple is the struct whose fields are its positions, so in hardware it is just wires, it flattens in waveforms as `t.0`, `t.1`, compares with ==, prints as `('1', 9)`, and matches position by position.

6.7 Arrays

implemented

Arrays have declared, directional index ranges (`Logic[7..0]` counts down, `Bit[4]` is `Bit[0..3]`), and indexing and slicing use the declared indices:

```
let data: Logic[31..0];
let byte: Logic[7..0] = data[7..0];      // MSB first, as written
let top: Logic[7..0] = data[..24];       // an open end takes the declared bound
let table: unsigned[8][4] = [10, 20, 30, 40];
```

```
y = table[sel];                          // a runtime index: a lookup table
matrix[row][col] = v;                    // every dimension may be runtime
```

- **Nested arrays read inside-out** when declared: each `[N]` wraps the whole type to its left, so `unsigned[8][2][3]` is three rows of `unsigned[8][2]`, indexed outer first, `m[row][col][bit]`. The rule keeps types substitutable: with `type Row = unsigned[8][2];`, `Row[3]` is the same type.
- **Slices and elements are places:** `y[3..0] = a` and `y[i] = b` write part of a signal and keep the rest; partial ranges (`a[..1]`, `a[2..]`) work on ports, signals, locals and constant tables alike.
- **Unconstrained arrays** leave their range to the use, as VHDL's `range <>`: `string = Char[]`, and a port `s: string[5]` supplies the length. In a testbench, `T[]` is a run-time array: `read<integer[]>(path)` sizes it to a file, `xs.push(v)` appends, and `integer[][]` nests.
- The system attributes `'length`, `'left`, `'right`, `'high`, `'low` and `'ascending` read an array's shape as elaboration-time constants, and a named `range` constant (`const BYTE: range = 7..0;`) works in type and slice position.

6.8 Newtypes and aliases

implemented

`type Word = unsigned[32];` is a transparent alias: `Word` **is** `unsigned[32]`. A new nominal type wraps an existing representation: `struct Meter(real);` and `enum Logic(ULogic);` share a representation but are distinct types, with the conversion between them generated automatically. Derivation never adds members or inherits behaviour; bigger types are built by composition. A newtype over an array forwards the operators its base's element provides, so a newtype over `unsigned[8]` keeps `+` and `==`.

6.9 Every value starts defined

implemented

A signal without an initializer starts at its type's default value, given by the `Default` trait: `'U'` for `Logic`, `'0'` for `Bit`, zero for numbers, the first variant for an enum, and each field's default for a struct. `T::default()` names it; a type that declares its own `default` (or an inherent `new`) runs that body. siox signals are always initialized; they may be undriven, and the compiler warns when they are.

7 Abstraction

7.1 Traits and generic implementations

implemented

A trait is a compile-time contract: a set of functions a type must provide. There are no vtables and no run-time dispatch; every call resolves when the design is elaborated.

```
trait Source<T> {
    fn send(value: T);
    fn can_send() -> Bit;
}

impl<T> Source<T> for Stream<T> Source {
    fn send(value: T) {
        self.valid = '1';
        self.data = value;
    }
    fn can_send() -> Bit { return self.ready; }
}
```

A member's receiver is implicit, as C++'s `this`: `fn send(value: T)` has one, named `self` in its body, and `static fn` marks a member without one, in traits and implementations alike (§ 11.6).

Generic implementations are written with Rust's binder, `impl<W: integer> Counter<W>`. Bounds restrict generic code to types that provide what it uses, and a blanket implementation over arrays lifts an element's behaviour to every array of it:

```
impl<T: Resolve> Resolve for T[] {
    fn resolve(rhs: T[]) -> T[] {
        let result: T[] = self;
        for i in self'range { result[i] = self[i].resolve(rhs[i]); }
        return result;
    }
}
```

7.2 Operators

implemented

Operators are traits, as in Rust. Each standard operator has its own: `Add`, `Sub`, `Mul`, `Div`, `Rem` (%, with the dividend's sign), `Shl`, `Shr`, `And`, `Or`, `Not` and `Neg` (unary `-`), with a method named after it. A type gets `+` by implementing `Add`, and the method can also be called directly, `a.add(b)`:

```
impl Add<Complex, Complex> for Complex {
    fn add(rhs: Complex) -> Complex {
        return Complex { .re = self.re + rhs.re, .im = self.im + rhs.im };
    }
}
```

The traits double as bounds: a generic function over any type that can be added says `T: Add<T, T>`.

Comparisons are two traits, as rustc's `PartialEq` and `PartialOrd`: `Eq` gives `==` and `!=`, `Ord` gives `<`, `<=`, `>` and `>=`, and every method returns `Bool`. A type with equality but no order implements `Eq` alone, and an order may leave values unordered, as IEEE floating point does for NaN:

```
impl Eq<Version> for Version {
    fn eq(rhs: Version) -> Bool {
        return self.major == rhs.major and self.minor == rhs.minor;
    }
}

impl Ord<Version> for Version {
    fn lt(rhs: Version) -> Bool {
        return self.major < rhs.major or (self.major == rhs.major and self.minor <
rhs.minor);
    }
    fn le(rhs: Version) -> Bool { return not rhs.lt(self); }
}
```

`abs`, `rem` and `mod` are functions, as in mathematics, and live in `std::math` with `min`, `max` and the constants: `abs(x)`, `rem(a, m)` (the dividend's sign) and `mod(a, m)` (the divisor's). They are generic, so one `abs` serves `integer`, `signed`, the fixed formats and `float` alike.

New operators are `CustomOperator`, parameterized by their symbol. Any symbol or word that is not reserved by the grammar can be an operator, and its binding power is bound inside its implementation; the standard operators' binding powers are the grammar's and cannot change:

```
impl CustomOperator<"xor", Logic, Logic> for Logic {
    attr precedence = 35;
    fn apply(rhs: Logic) -> Logic { … }
}
```

That is how `xor`, `nand`, `nor` and `xnor` exist: the standard library declares them; the compiler does not know them. A custom operator's result is the type its implementation declares, width, range and alias included, so `(a clip b) xnor c` uses the declared `unsigned[3]` of `clip`, not the width of its inputs.

An operator always dispatches on its operands' own types: a value is never converted to fit an operator, so `u + arr` with an `unsigned[8]` and a `Logic[8]` is an error whose help writes the conversion out.

7.3 Literal prefixes and suffixes

implemented

`10ns`, `100MHz` and `5i` are ordinary literals with a suffix, and the suffix is a trait implementation in source: `impl Suffix<"ns", integer> for time` builds a `time` from the number. Bit-string prefixes work the same way through `Prefix`. A library can add its own units and literal forms.

7.4 Conversions: `From`, `Into` and `Default`

implemented

Three traits, three jobs:

Trait	Written	Converts
`From<S> for T`	`T(x)`	explicitly, at a conversion: `Complex(10)`, `Logic(u)`, `ufixed<8, 4>(x)`
`Into<T> for S`	nothing	implicitly, where an `S` is stored as a `T`: a `let`, an assignment, an argument, a returned value, a struct-literal field
`Default for T`	`T::default()`	nothing; it is the value a signal starts at

A nominal vector family and a plain array of the same elements are different types, joined only by `Into`. The standard library declares one both ways between `Logic[]` and each family that is a newtype over it (`unsigned`, `signed`, `ufixed`, `sfixed`, `float`), as a copy by position, so a logic vector read from a port stores straight into an `unsigned[8]` or a `ufixed<8, 4>`. Conversions that scale, round or saturate (between two fixed formats, from `ufixed` to `unsigned`) stay an explicit `T(x)`. An `Into` is declared by itself and needs no `From`; unlike Rust's, a `From` never implies one, so every implicit conversion is one a library chose to allow. A composite value converts to a vector or a number only through a `From` declared for it.

Types that are not built-in arrays can still be indexed by implementing `Index` and `IndexAssign`.

7.5 `#[derive]`

implemented

`#[derive(Eq, Ord)]` before a struct writes the comparison impls a user would write by hand: == field by field, and </<= lexicographically in declared order, with `!=`, > and >= from the traits' defaults.

```
#[derive(Eq, Ord)]
struct Version { pub major: unsigned[4], pub minor: unsigned[4] }

if v1 < v2 { … }      // major first, then minor
```

Generic structs derive for every instance (`impl<A, B> Eq<Pair<A, B>> for Pair<A, B>`), and a newtype compares as its base. The impls belong to the struct's own module, and an error inside one (a field whose type has no ==) is shown in the written-out impl, noted with the derive that wrote it. `Resolve` does not derive, because a struct signal already resolves field by field.

7.6 Type attributes

implemented

An implementation may declare an attribute that every value of its type answers, written in terms of `self`'s shape:

```
impl ufixed {
    pub attr integers: integer = self'high + 1;    // x'integers
    pub attr fractions: integer = 0 - self'low;    // x'fractions
}
```

A type attribute is a constant, as `x'high` is: it reads only shape (system attributes, constants and functions over them), never the value. It is how the library states the per-type facts VHDL leaves to `'high`/`'low` arithmetic, and a format reads its target's through a qualified call, `ufixed<6, 2>::resize(x)`.

7.7 Functions

implemented

In hardware a function is combinational logic: it is inlined where it is called, and evaluated at compile time when its arguments are constants, so `unsigned[clog2(DEPTH)]` works as a width. Generic functions bind their parameters from their arguments, including integer ones: `fn widen<W: integer>(x: unsigned[W]) -> unsigned[W + 4]` called with an `unsigned[4]` is instantiated with `W = 4`, as rustc monomorphizes.

A member has an implicit receiver and is called on a value (`x.f()`); a `static fn` has none and is called through its type's path (`Unicode::code(c)`, `Entity::helper(x)`, `T<args>::f(x)`). In a testbench a function is an ordinary procedure: it may loop, `await`, return early, write its caller's places through its parameters (`std::mem`'s `swap` and `replace`), and recurse, each recursive call keeping its own frame.

`#[inline]` chooses whether the simulation expands calls or shares one native copy; it never changes what a call means. A function may keep state between calls in a `static let`: one copy per function, as C's `static` (§ 11.6). C functions are available through `extern "C"` blocks (§ 10.4).

8 Metadata and directives

siox separates two things other HDLs mix: **metadata**, which only tools read, and **directives**, which change what the compiler does.

	Spelling	Removing it changes
metadata	`attr keep for probe = true;`	only what a downstream tool sees
directive	`#[test]`, `#[allow(...)]`	what the compiler emits, accepts, or reports

8.1 Declared attributes

implemented

An attribute is declared once, with a type, the targets it may apply to, and a default. It is then bound from outside the declaration it describes, and read back with the tick:

```
pub attr external_clock: Bool for Pll = false;   // declaration
pub attr speed: integer for entity, let = 1;

impl Top {
    attr external_clock for p = true;           // binds the instance `p`
    attr speed = 3;                             // binds the entity `Top`

    let p: Pll = { .clk = clk, .locked = l };

    count = p'speed;    // the binding on p, else Pll's binding, else the default
}
```

An implementation can also declare an attribute that every value of its type answers, such as the fixed-point formats' `x'integers` and `x'fractions` (type attributes, in the previous chapter).

Because every attribute has a default, a read always has a value. Bindings are compile-time constants, so a read costs nothing in the circuit. Binding an attribute twice on one declaration is an error; a binding never silently overrides another. Metadata bound on an instance appears on it in the elaborated instance tree (`p: Pll [external_clock = true]`), for netlist and constraint tools to export.

8.2 Directives

implemented

`#[...]` is reserved for directives. Like rustc's built-in attributes they are part of the compiler and declared nowhere: they cannot be imported, renamed or shadowed, and take no value. A module's own `attr test` is metadata, bound with `attr test for X = …;`, and never makes a test.

- `#[test]` compiles an entity into the test executable.
- `#[allow(lint, …)]`, `#[warn]`, `#[deny]` and `#[forbid]` set lint levels.
- `#[inline(always)]` and `#[inline(never)]` choose whether the simulation expands a function's calls in place or calls one shared copy; `#[inline]` is a hint to expand. Without one, small functions are expanded and larger ones shared. It is a choice about the simulation only: a function means the same in hardware either way.
- `#[derive(Eq, Ord)]` on a struct writes its comparison impls.
- `#[latched]` and `#[latch]` will make a function a pipeline and mark its stages (§ 11.1) proposal.

There is deliberately no `#[cfg]`. The compile-time target is a value, `std::target`, folded before reachability: both branches of a choice are type-checked and there is one IR, where a directive that deletes code would let two configurations drift apart.

Metadata written as `#[...]` is an error whose help gives the `attr` binding that replaces it.

8.3 Lints

implemented

Every warning is a **lint** with a snake_case name, as in rustc: `possible_latch`, `unused_signal`, `undriven_output`, `combinational_loop`, `dead_assignment`, `non_exhaustive_match`, `incomplete_struct_literal`, and the others. A level is set for the item or statement a directive precedes, for a whole module with `#![...]`, or for everything from the command line with `-A`, `-W`, `-D` and `-F`:

```
module design;
#![deny(warnings)]                    // every warning in this module fails the
build

impl Pipeline {
    #[allow(undriven_output)]         // a probe kept for the waveform
    let probe: unsigned[8];
}
```

The innermost level wins; `forbid` cannot be lowered by anything inside it. A denied lint is a real error, and says which directive or flag made it one.

9 Verification

9.1 Testbenches are entities

implemented

A testbench is an entity marked `#[test]`. It has no ports; it instantiates the design under test, drives it from processes, and checks it. Testbench processes may do what hardware cannot: wait for time to pass, write files, draw random numbers, print.

```
#[test]
entity AdderTest {}

impl AdderTest {
    let a: unsigned[8];
    let b: unsigned[8];
    let sum: unsigned[8];
    let dut: Adder = { .a = a, .b = b, .sum = sum };

    stimulus: process {
        seed(42);
        for i in 1..100 {
            a = unsigned[8](randint(0, 255));
            b = unsigned[8](randint(0, 255));
            await 1ns;
            assert!(sum == a + b, "the adder wraps like unsigned[8]");
        }
        print!("checked {} random additions", 100);
    }
}
```

A testbench may have several processes, each waiting on time or on signals by itself; the scheduler interleaves them by simulation time and delta cycle, so a monitor and a stimulus process run side by side, and none runs to completion before the others start. Here the counter of the first example gets a stimulus that resets it twice and a monitor that reports every edge:

```
impl CounterTest {
    // clk, rst, count and dut as in the first example

    clock: process {
        clk = not clk after 5ns;            // runs each time clk changes
    }

    stimulus: process {
        await 10ns; rst = '0';              // release reset
        await 30ns; rst = '1';              // reset again for one edge
        await 10ns; rst = '0';
        await 20ns;
    }

    monitor: process {
        for i in 0..7 {
            await clk.rising();
            print!("count = {}", count);    // after this edge has settled
        }
    }
}
```

Figure 5 shows the run: the signals from its VCD, and beneath them when each process ran. Every process starts at 0. The clock process runs at each change of `clk`; the stimulus runs only when its `await 10ns` or `await 30ns` expires; the monitor runs at each rising edge, and an `await` in a testbench resumes only after the edge's consequences have settled, so it prints the new count (0, 1, 2, 3, then 0 after the second reset).

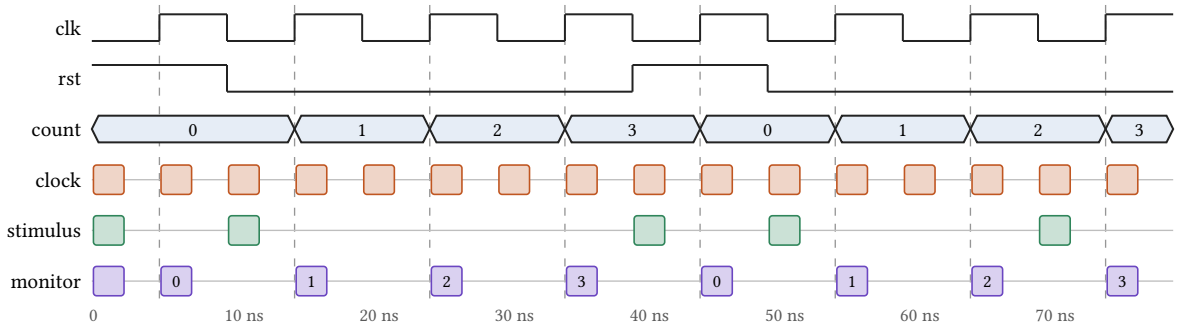


Figure 5: `CounterTest` with three processes, simulated. The upper rows are the VCD; each lower row marks the moments a process ran, between which it was suspended in an `await`. The monitor's marks carry the count it printed. All three share one host thread; at a shared time they run in a fixed order, so a run always reproduces.

Several testbenches may live in one file. Each is named by its qualified path (`adder::AdderTest`) and runs independently.

9.2 The test executable

implemented

`sioxc --test file.siox -o tests` compiles every `#[test]` entity in the file into one native executable, as `rustc --test` does. The compiler never runs it. The executable:

- runs every test, or those whose name matches a filter (`./tests adder::AdderTest`);
- reports each test as passed or failed, with the source location of a failed assertion, and counts warnings;
- writes waveforms when asked: `./tests -o run.vcd` (text VCD) or `./tests -o run.fst` (compressed FST), or both in one run. Several tests share one monotonic timeline.

9.3 Checking and reporting

implemented

- `assert!(cond, "message")` fails the test when `cond` is false.
- `warn!(cond, "message")` reports and counts, and the test still passes.
- `print!("x = {}", x)` formats a line. Enum and logic values print symbolically (`Idle`, `'Z'`), characters and strings as text, and numbers in full, however wide. Placeholders take Rust's specs: `{:.3}` rounds to three decimals, `{:.2e}` is scientific, `{:#x}` hexadecimal, `{:>8}` pads and aligns.
- Structs print as `Packet { kind: Data, len: 12 }`, tuples as `('1', 9)`, arrays as `[1, 2, 3]` and logic vectors as `10XZ`; `{:?}` gives Rust's debug form. A type prints its own way by implementing `Display`, whose `fmt` body uses `write!`; `float`, the fixed-point formats and `Complex` do.
- `stop()` ends the test as passed so far; `finish()` ends the simulation.
- A value that leaves a ranged type, or a file read that fails, is reported with the signal's path or the source location.

The macros capture their source location, which is why they are macros: a failure names the line of the call, even when the call is inside another macro. `error!("message")` fails unconditionally, like Rust's `panic!`. All four are declared in `core` (§ 11.2).

9.4 Stimulus services

implemented

- **Randomness** is deterministic: `rand()`, `randint(lo, hi)`, `uniform()` and `seed(n)` come from one generator with a fixed default seed, so a run always reproduces.
- **Files**: `read<string>(path)` decodes UTF-8 text, `read<integer>(path)` and `read<unsigned[16]>(path)` read binary fixtures, `read<integer[]>(path)` sizes a run-time array to the file, and `exists(path)` probes for one. In hardware an initializer read happens at compile time, baking a ROM image; in a testbench it happens when the test runs.
- **Run-time arrays**: a testbench `T[]` grows with `xs.push(v)` and nests (`integer[][]`), for scoreboards and expected-value queues.

9.5 Waveforms and debugging

implemented

Waveforms record every signal with its hierarchy path: instances and labelled generate scopes nest (`top.stages[0].s.o`), enums appear by name, `Logic` appears with `x` and `z`, and structs and arrays flatten to one trace per field or element. The files open in GTKWave or Surfer.

A failure names its source without a debugger: a failing assertion, a value leaving its range or a failed file read prints `--> file:line:col` with the source line and a caret. For stepping, `sioxc --test -g` emits DWARF for the siox source with optimization off, so GDB sets breakpoints on siox lines, steps with `next`, and prints state by its elaborated path (`FifoTest::d::count`), wide vectors, structs and directed arrays included. A debug build simulates exactly as an ordinary one; the waveforms are byte-identical.

10 The compiler

10.1 One pipeline

implemented

`sioxc` is one Rust crate: a library of stages and a thin command-line driver. Each stage hands the next a complete product, and each stage keeps going after an error so a single run reports as much as it can.

Stage	What it does
parse	Tokens and a syntax tree for every module the program imports, transitively.
expand	User macros expanded where they are called, and `#[derive]` impls written into their struct's module.
attributes	Attach `attr` bindings to their targets and fold attribute reads into values; collect lint levels.
resolve	Every name to the declaration it means: modules, imports, visibility, paths.
type-check	Types, widths, directions, trait bounds, operator dispatch, conversions.
elaborate	Parameters substituted and generation unrolled into a concrete instance tree.
lower	A digital IR: signals, combinational drivers, event-controlled updates, and one control-flow form for every process.
codegen	LLVM, then a native object or test executable linked with a fixed runtime.

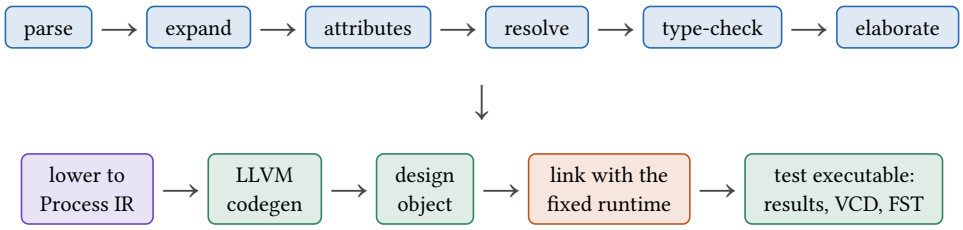


Figure 6: The pipeline. The front end produces one checked instance tree; hardware and testbench processes alike become Process IR; the native code links with a runtime that is the same for every design.

The digital IR keeps one distinction from VHDL's model at its centre: a **driver** (a wire: `target = expr` under a condition) and an **event block** (`on condition: next(target) = expr`) are different things, and `'event` and `'old` are first-class operations. Hardware processes and testbench processes both converge on one control-flow representation, validated before code generation, so there is one path from source to executable and one meaning.

10.2 Native simulation

implemented

The compiler does not interpret designs. It emits the design as native code and links it with a small runtime, fixed and independent of the design, that schedules processes in delta cycles, advances simulation time, applies delayed writes with inertial semantics, and writes VCD and FST. A simulation is therefore a native program, with no interpreter in the loop.

10.3 Diagnostics

implemented

Every diagnostic has a stable code (`E-P014` is a conflicting driver, `W-P002` a possible latch), a primary location, labelled secondary locations, notes and a suggested fix. Migration errors say exactly what to write instead:

```
error: `using` was split into `use` and `type`; this is a type alias
  --> design.siox:3:1
   = help: write `pub type Word = unsigned[32];` (`type`)
```

Warnings are lints whose levels are set in source or on the command line (§ 8.3). The compiler prefers no error over a wrong one: where it cannot yet decide something soundly, it stays silent.

10.4 C functions

implemented

C libraries connect through extern blocks, as in Rust:

```
extern "C" {
    pub fn cbrt(x: real) -> real;      // from the C math library
    pub fn labs(v: integer) -> integer;
}
```

`real` crosses as `double`, `integer` as a 64-bit signed word, packed numeric values up to 64 bits as an unsigned word, and `std::ffi`'s `c_int`, `c_short`, `c_float` and the others at their C widths. A function without a return type is C `void`, a statement a testbench runs in order. A struct or fixed array of C scalars is passed by pointer to a C-laid-out copy, which the caller receives back (an out-parameter); wider values cross as little-endian words. Calls work in hardware expressions, clocked updates and testbenches, and a test executable links any object given with `--link`. The standard library does not use C: its `real` math (`sqrt`, `sin`, ...) is compiler intrinsics, so a design never depends on a host library it did not ask for.

10.5 Embedding and editors

implemented

The compiler is a library first. `siox::compiler` takes a request (a file or an unsaved editor buffer, and the artifact wanted) and returns every product it produced: modules, resolution, types, the instance tree, the IR, and the diagnostics. The command-line driver is a thin adapter over it, and the `siox-lsp` language server, in its own repository, uses the same library for live diagnostics, go-to-definition, hover, completion and rename, so editor and compiler never disagree.

Inspection outputs show each stage: `--emit tokens`, `ast`, `source` (the canonical printed form, also the migration tool), `expanded` (after macros and derives), `tree` (the instance hierarchy), `metadata`, `ir` and `llvm-ir`. `-g` adds DWARF so GDB steps the siox source (see Verification).

Part III · Where siox is going

11 The digital language, growing

Phase 1 is complete: an audit in October 2026 traced every requirement, deliverable and example to the compiler, the standard library and the test corpus (`docs/phase1-audit.md` in this paper's repository). It is a baseline, not a finished language. This chapter describes what is designed and written down as proposals, in roughly the order it is expected to land. Each proposal lives in the compiler repository under `docs/proposals/` until it is built; its substance then moves into the reference documents beside this paper (`docs/`, starting with the language specification) and the proposal is deleted. The sections below give every proposal as it stands at the end of October 2026, including the ones the owner has paused and the open questions still to be decided.

11.1 Pipelined functions

proposal

A function is combinational: one clock cycle, however deep its arithmetic. A pipelined datapath today is an entity written by hand, with a register per stage per value and the latency known only to its author. siox adopts Spade's answer: stage boundaries are part of the source, the compiler carries values across them, and latency is checked at every use. It spells them with two directives rather than new keywords, because a stage boundary is not new computation but an instruction about when results are kept.

```
#[latched]
fn mac(clk: Bit, a: signed[16], b: signed[16], acc: signed[32]) -> signed[32] {
    #[latch] {                                  // stage 1
        let product: signed[32] = signed[32](sext(a)) * signed[32](sext(b));
    }
    #[latch]                                    // stage 2
    let sum: signed[32] = product + acc;
    return sum;
}

impl Filter {
    #[latched(2)]
    y = mac(clk, x, coefficient, offset);       // mac's result, two cycles later
}
```

- `#[latched]` makes a function a pipeline; its first parameter is the clock.
- `#[latch]` marks one stage: on a block, everything in the block; without braces, the one statement that follows. At the end of a stage every value it computed or carried is registered, and a `let` inside a stage block stays visible after it, one cycle later: a stage is a step in time, not a scope.
- The depth is the number of stages, stated by `#[latched(N)]` and checked against the body. Every call site states the depth it expects too, so changing a pipeline's depth breaks its users at compile time instead of silently shifting their timing by a cycle.
- `#[latch(name = fetch)]` names a stage, and `stage(fetch).x` or `stage(-1).x` reads a value in another stage, for forwarding. Using a value before the stage that computes it is an error that says how many stages early it is.
- `#[latch(enable = cond)]` stalls a stage while `cond` is false, and a stall propagates to every earlier stage. `stage'ready` and `stage'valid` say whether a stage accepts input and whether its contents are real; valid bits start false, because every siox signal starts at its default.

Figure 7 shows what the call site's `#[latched(2)]` promises: each input reaches `y` two clock edges after it is presented, and one new input enters every cycle.

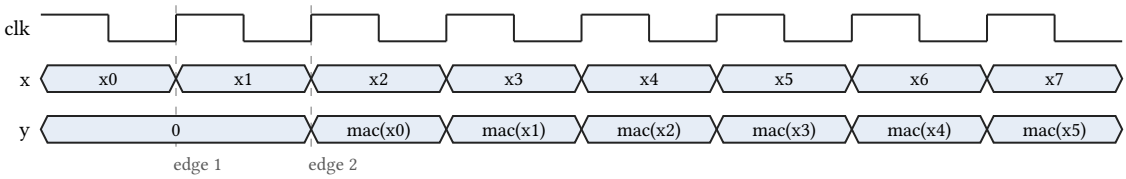


Figure 7: The intended timing of `y = mac(…)` with `#[latched(2)]` (a proposal; not simulated). Before the first result arrives, `y` holds its type's default.

The pipeline registers are ordinary registers in the IR and in the eventual RTL; no backend needs to understand the directives. One naming question is open: in hardware a **latch** is a level-sensitive element, while these stages are edge-triggered registers, so `#[pipelined]`/`#[stage]` are possible alternative names.

11.2 Macros

implemented

Some abstractions generate structure rather than compute values. Macros give siox a hygienic, declarative way to write them, invoked with `!` so they stay visibly distinct from calls:

```
pub macro debug_signal($name: ident, $ty: type) {
    #[allow(unused_signal)]
    let $name: $ty;
}

impl Cpu {
    debug_signal!(probe, unsigned[32]);
}
```

Implemented, following Rust's declarative macros 2.0:

- Parameters bind syntax fragments (`expr`, `ident`, `type`, `path`, `stmt`, `item`, `tokens`), not values, and a macro may have several forms, chosen by the arguments.
- A call expands to an expression, statements, implementation members or module items, depending on where it stands.
- Expansion happens before name resolution, so generated code is checked exactly as if it had been written by hand.
- Hygiene: names a macro declares cannot capture the caller's names, and the body's other names mean what they mean where the macro is declared.
- Macros are items, imported with `use`; `--emit expanded` shows what hardware a macro produced.
- Repetition unrolls at expansion time: a `$xs: expr...` parameter takes the remaining arguments, and `for macro $x in $xs join and { $x }` repeats over them.
- An assertion that fails inside a macro reports the line of the call.
- The built-in `assert!`, `warn!`, `print!` and the new fatal `error!` are ordinary macros declared in `core`, each over a compiler primitive that only `core` may use:

```
pub macro assert($cond: expr, $rest: expr...) { builtin # assert($cond, $rest) }
pub macro error($rest: expr...) { builtin # assert(false, $rest) }
```

Still to come is tooling rather than language: a diagnostic inside a nested expansion should name every invocation that produced it, and editors should be able to show generated declarations next to the call that

made them. Not proposed: procedural token-stream macros, user-defined `#[...]` attribute macros, and arbitrary code execution at compile time.

11.3 `core` and `std`

implemented

The library is split along one question: **could an external library have written this?**

	Contents
`core`	What only the compiler can provide, reached through the language: the kernel types (`integer`, `real`, `Char`, `Bool`, `string`, `Range`); the hook traits the compiler calls (`Add`, `Rem` and the other operators, `CustomOperator`, `Eq`, `Ord`, `Match`, `Prefix`, `Suffix`, `Index`, `Boolean`, `Resolve`, `Default`, `From`, `Into`, `Display`, `LogicEncoding`); the intrinsics behind `real` math; the macros, including a new fatal `error!`; and the simulator services (`await`, `stop`, `finish`, files, randomness). Compiled into `sioxc`, so it always matches the compiler.
`std`	Everything a user programs with: `Bit`, `ULogic`, `Logic` and their truth tables, `unsigned`/`signed` and conversions, ranged integers, `Complex` and math, text encodings, time and frequency, fixed and floating point, and, to come, vectors and matrices.

Implemented: `core` is compiled into `sioxc` and laid out like rustc's: `core::primitive` (`Bool`, `string`, and `integer`'s methods), `core::ops`, `core::cmp` (`Eq`, `Ord`, `Ordering`, `Match`), `core::convert` (`From`, `Into`), `core::default` (`Default`), `core::fmt` (`Display`) and `core::macros`; `std` re-exports each module under the same name. The directives are not declared at all: like rustc's, they are built into the compiler. Each `core` declaration tells the compiler its role with a lang item, as rustc's `#[lang = "add"]` does:

```
attr lang for Add = "add";
```

The compiler finds its hooks by role and never by path, and only `core` and `std` may bind `lang`, so a user trait named `Boolean` stays an ordinary trait. The built-in macros are `core` declarations too (§ 11.2). The `std` content still to come is listed in the next section.

Metadata follows the same question. `std::attrs` declares only what nearly every flow needs to know: `keep`, `top`, `clock`, and a foreign entity's `library` and `name`. Vendor settings (RAM styles, FSM encodings, pin assignments, I/O standards) belong to vendor packages, namespaced as `attr vivado::ram_style for …`, so the language never grows a vendor's vocabulary.

11.4 The standard library

partly implemented

std is the mandatory, vendor-independent base: data types, conversions and small helpers every technology has. Memories, FIFOs, stream adapters and verification components are IP, and belong to vendor packages and libraries. Each piece comes with documentation and a runnable example:

1. **Synchronizers and reset helpers** implemented: `std::sync`'s two-flop synchronizer, reset synchronizer, edge detector and pulse synchronizer.
2. **Fixed point** implemented: `std::fixed`, after VHDL's `fixed_pkg`, with the format as type parameters: `ufixed<8, 4>` has eight bits, four of them fraction; arithmetic and division keep the format, and the format's own constructor converts a number, `ufixed<8, 4>(2.5)`, or resizes another format, `ufixed<12, 6>(x)`, rounding and saturating.
3. **Floating point** partly implemented: `std::float`, after VHDL's `float_pkg`: `float<32, 23>` is IEEE-754 binary32 (32 bits, 23 of them mantissa), with addition, subtraction, multiplication, division, square root, IEEE comparison (a NaN is unordered), conversions to and from the fixed formats, and the

constructor `float<32, 23>(1.5)`, rounding to nearest even, in processes and in hardware entities alike. Still to come: subnormals (flushed to zero today) and the other rounding modes.

4. **Complex numbers** implemented: `std::math::Complex`, after `ieee.math_complex`, with `+`, `-` and the `i` suffix (`3.0 + 4.0i`).
5. **Linear algebra** (optional): `Vector<T, N>` and `Matrix<T, R, C>` over any element with the needed operator traits, as ordinary array-backed library types, once a design needs them.

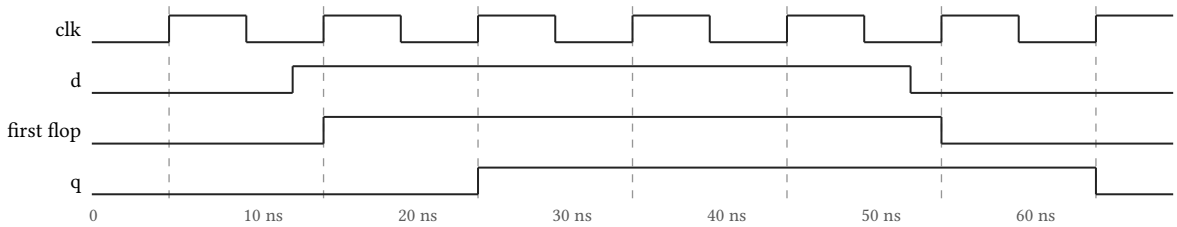


Figure 8: `std::sync::Sync2` in simulation: an input that changes between clock edges (13 ns) is taken by the first flop on the next edge and reaches `q` one edge later, so `q` only ever changes on an edge of `clk`.

11.5 Entity methods

partly implemented

An entity's interface is its ports, and a protocol between two entities (offer a value, learn whether it was taken) is today spread over several ports, declared twice and wired at every instantiation. Entity methods let a component expose that protocol once, as a function whose calls **elaborate into** ports, following Bluespec's method-to-port mapping:

```
impl Fifo {
    pub fn write(value: unsigned[8]) -> Bool { … }
}

impl Producer {
    if clk.rising() {
        if f.write(value) { value = value + 11; }
    }
}
```

A call materialises an enable port, one port per argument and one for the result; nothing is hidden, and the generated ports cost exactly what hand-written ones do. The first version allows one call site per method per instance, so no arbitration is needed. A call through an instance array works the same way: `stage[sel].write(v)` gives every element its own enable, gated by `sel`, so each still has one caller.

Associated functions without a receiver are implemented. Accessors and action methods are on hold: the owner paused them in October 2026, because an entity's body names its ports and state without a receiver, and a `self` parameter on an entity method would look unlike everything else in that body. The implicit-receiver proposal below answers that objection; whether the two land together is the proposal's open question.

11.6 Implicit receivers and `static`

partly implemented

A keyword, not a `self` parameter, tells a method from an associated function, the same for structs, enums, entities and views:

Declared in an implementation	Receiver
`fn f(...)`, `pub fn f(...)`	implicit `self`
`static fn f(...)`, `pub static fn f(...)`	none: called as `Type::f(...)`

```
impl Counter {
    pub fn increment() { value = value + 1; }         // fields by name
    pub fn set(value: integer) { self.value = value; } // `self.` when shadowed
    pub static fn from_count(value: integer) -> Counter {
        return Counter { .value = value };
    }
}
```

`self` stays as an expression and `Self` as the implementing type; `pub` controls only visibility. Constructors and conversions become explicitly static (`static fn default() -> Self`, `static fn from(value: S) -> Self`). The migration is loud: a receiver-free `fn` in an implementation used to be an associated function and now has a receiver, so every path call to it, and an impl of a trait's `static` member, fails with a message saying to write `static fn`, rather than changing meaning. A written `self` parameter, the old spelling, still compiles with a warning to remove it.

The owner accepted the model on 31 October 2026, and it landed in November 2026 across core, std and the corpus. Two parts follow: bare field names (`value` for `self.value`, shown above) and public entity methods.

The proposal also adds `static let`, state that outlives one activation of a process or one call of a function. Its rules:

- It is allowed in processes and functions, directly in the body, and an error in an entity body, where every `let` is already declared once. It cannot hold an entity, and its initializer is a constant, run once.
- It has **variable** semantics, as a VHDL process variable: a write takes effect at once, so the next statement reads it. That is what lets two calls in one process chain, and a counter function return 1, 2, 3.
- In a function there is one copy per function, as with C's `static` (decided November 2026); a generic function has one per specialization. A testbench's calls share it in order. In hardware it is one register: several calls in one process chain, as a VHDL function reused in a process does, while updates from separate processes (including two instances of the same entity) are a conflicting driver.

Testbench `static let` is implemented (November 2026); in hardware it is reported until its register lowering lands. It overlaps scoped blocks (below): `static let` is the lighter form for state one process or function owns, a block for state several processes share.

11.7 Scoped blocks

proposal

A large implementation collects all its state in one namespace. A `{ … }` block at implementation scope groups members under their own scope, with an optional label:

```
impl Cpu {
    counter: {
        let n: unsigned[8] = 0;          // private to the block
        pub let wrapped: Bit = '0';      // visible to the whole implementation
        step: process {
            if clk.rising() {
                n = n + 1;
                wrapped = if n == 255 { '1' } else { '0' };
            }
```

```
        }
    }
    overflow = counter.wrapped;          // a pub member, through the label
}
```

Names follow Rust's block scoping; a private member from outside is an error naming the block. `pub` exports one level and never past the entity, whose interface stays its ports. A labelled block is a scope in the hierarchy, as a generate label is (`counter.n` in the tree, waveforms and debugger); an unlabelled one adds no level. Blocks add no semantics: elaboration flattens them, and no register is ever inferred from a local.

11.8 Payload enums

on hold

Variants that carry data, as in Rust, for optional and fallible results, protocol messages and state machines with per-state data:

```
pub enum Option<T> { None, Some(T) }
enum Command {
    Idle,
    Read(unsigned[16]),
    Write { addr: unsigned[16], data: unsigned[32] },
}

match c {
    Command::Idle => ready = '1',
    Command::Read(addr) => q = memory[addr],      // binds `addr`
    Command::Write { addr, data } => memory[addr] = data,
}
```

A payload enum is a tag plus a payload region as wide as its largest variant, the variants overlaid as in a SystemVerilog tagged union; a field is in scope only inside its variant's arm, so another variant's bits are never read. Patterns that bind (`Some(x)`) are the first of their kind and the largest piece of work; the match coverage proofs that exist already extend to variants. std would gain `Option`, `Result` and `checked_add`.

The rule agreed in October 2026 keeps it apart from the logic system. An enum is either a **value enum**, today's kind, whose variants may be characters or carry explicit discriminants (`'U' = 4`, `Add = 0`), or a **sum type**, whose variants may carry data and whose tag the compiler chooses. A character or numbered variant has no payload, as in Rust a variant with an explicit discriminant has none, so `Bit`, `ULogic`, `Logic` and every character enum keep their representation. A character literal inside a payload takes its type from context (`let b: Option<Bit> = Some('1');`). The owner has paused the proposal; the open questions are what unused payload bits hold (`'0'` or `'-'`), a checked accessor, and dropping the tag for a niche, as Rust's `Option<&T>` does.

11.9 Binary and unary custom operators

proposal

`CustomOperator` is binary only. The proposal splits it into two contracts with explicit arity, so prefix operators can be declared too:

```
impl BinaryOperator<"clip", unsigned, unsigned[3]> for unsigned {
    attr precedence = 35;
    fn apply(rhs: unsigned) -> unsigned[3] { return unsigned[3](self + rhs); }
}
impl UnaryOperator<"flip", Logic> for Logic {
```

```
    fn apply() -> Logic { … }                    // `flip a`
}
```

The declared result keeps its width, range and alias identity through a chain, which `CustomOperator` already does. The standard operator traits (`Add`, `Not`, `Neg`, …) and the comparisons are unchanged, and an old declaration fails with a migration message instead of being reinterpreted.

11.10 The last implicit conversion site

partly implemented

`Into` converts at a `let`, an assignment, an argument, a returned value and a struct-literal field. A port connection is the remaining site: connecting an `unsigned[8]` signal to a `Logic[8]` port should convert as storing it does. It waits for the checker to type a connection against its port, which is in progress.

11.11 Compiler foundations

partly implemented

Changes borrowed from rustc's architecture, each independent. Lang items (§ 11.3) have landed: `core` marks the declarations the compiler hooks into, and the compiler finds them by role. The rest remain:

1. **UI tests**: every diagnostic pinned by a snapshot test, with inline `//~ ERROR` annotations and a `--bless` mode.
2. **Diagnostics for tools**: `--explain E-P014`, JSON output, and fix suggestions that a `--fix` mode can apply, making every future syntax migration one command.
3. **One constant evaluator** shared by every stage, replacing more than a dozen that grew separately, several of them added by bug fixes that each needed constants in one more place.
4. **A resolved tree** between type checking and elaboration, so later stages work with declaration identities and registries rather than re-reading the syntax.
5. **No name lookups after resolution**: the last few hooks still found by spelling move to lang items and resolved identities.

None of the five has landed yet (October 2026); each lands on its own.

11.12 Parallel simulation

proposal

A design has many processes, and today one host thread runs them all: the fixed runtime acts like a small cooperative scheduler, running a process until it suspends, settles or finishes, then the next. The proposal keeps that model and adds a bounded pool of worker threads that run, within one delta cycle, only process slices proven independent of each other. Effects are isolated and merged in a fixed order, so a run with any number of threads produces the same results, diagnostics and waveforms as a run with one; anything not proven independent falls back to serial execution. The thread count is an option of the test executable (`--threads N`), not of the compiler, and the same Process IR, native entries and runtime serve both modes. No new language construct is needed.

The unit of work is an **epoch**: the processes ready at one simulation time and publication boundary. On one thread they run one after another in a fixed reference order, process by process. With workers, the coordinator splits that ordered list into batches that may run side by side, and anything with an effect it cannot isolate (a file read, a random draw, a print, a write another process reads straight away) runs alone in its place in the order (Figure 9). Speed comes only from the batches; the order of the results never changes.

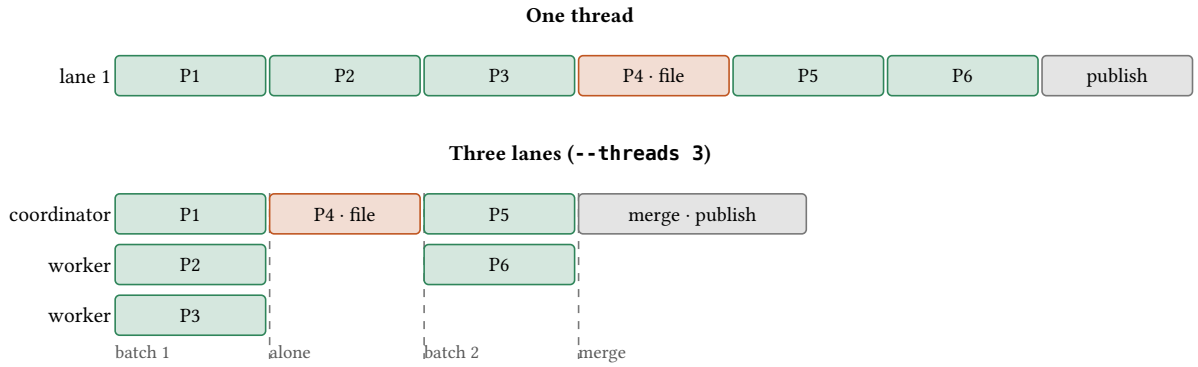


Figure 9: One epoch on one thread and on three lanes. P4 reads a file, so it runs alone, between the batches before and after it. The coordinator merges every slice's effects in process order before publishing, so both runs end in the same state.

Whether two slices may share a batch is decided before they run, from effect summaries the compiler derives from each process's control-flow graph: which signals it reads, which storage it writes, what it schedules, and whether it touches the host (Figure 10). Anything unknown counts as a conflict.

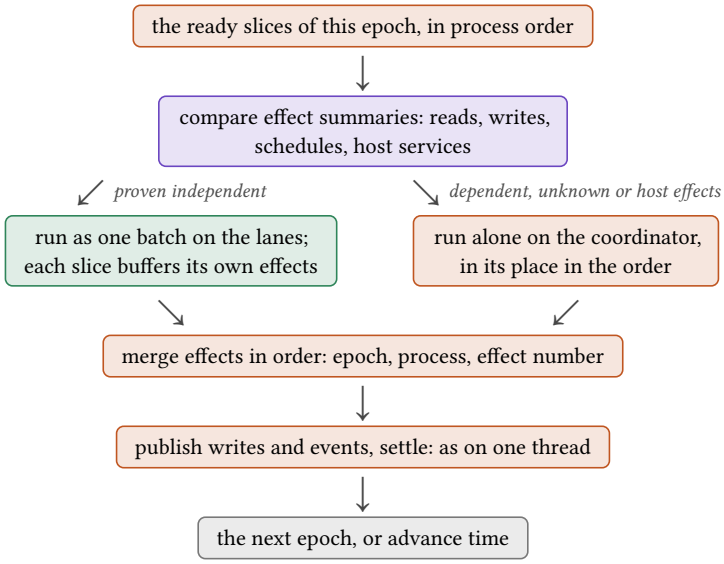


Figure 10: How the proposed runtime dispatches an epoch. Only slices proven independent reach the workers; everything else keeps its serial place. The merge and publication steps are the ones a single thread already performs.

11.13 Co-simulation and the simulator interfaces

proposal

Today a siox design runs only in its own executable. Two things a hardware team needs go through the same two IEEE interfaces in opposite directions: VPI (Verilog and SystemVerilog) and VHPI (VHDL).

- **siox as a client:** `sioxc` builds a siox top as a shared library plus a wrapper module or entity in Verilog and VHDL. A host simulator instantiates the wrapper like any other block, and the library, loaded as its VPI or VHPI plugin, steps the siox runtime from the host's callbacks. VHDL first, on nvc; Verilog second, on Icarus.
- **siox as a server:** the siox executable implements VPI, then VHPI, so external tools attach to it. cocotb is the first client, for Python testbenches.

Both rest on one refactor: the test runner's loop split into steps an outside scheduler can drive (start, apply inputs, settle, advance time, read outputs), with the `--test` executable as one driver of it. Values crossing follow the interoperability rules: logic is exact through VHPI and normalized to `X01Z` for four-state VPI. No language change is needed. The cocotb work was deferred in October 2026, when the API work closed; the proposal stays the design for when it resumes, and the same steppable runtime will serve the mixed-signal coordination of Phase 2.

12 Phase 2: analogue and mixed signal

Phase 2

The second phase adds continuous behaviour to the language without weakening the digital one. A board's power supply, filters, sensors, amplifiers and clocks are as much part of the circuit as its logic, and siox aims to describe them in the same language and simulate them together.

12.1 What it will do

- **Domains and quantities.** An analogue model works with continuous quantities in physical domains: voltage and current in the electrical domain, and later others (thermal, mechanical). A domain defines what is measured **across** a connection (a potential, such as voltage) and what flows **through** it (a flow, such as current).
- **Equations, not assignments.** Analogue behaviour is a set of relations that must hold at every instant: a resistor states that the voltage across it is its resistance times the current through it; a capacitor relates current to the rate of change of voltage. Time derivatives and integrals are first-class (`'ddt` is already reserved, and rejected by the digital compiler today).
- **Solvers.** The compiler collects the equations of a whole network into a system, and a solver integrates it over time. Solver choice, tolerances and step control are part of the model, and a solver that fails to converge points back at the source equations and domains, not at matrix indices.
- **Bridges.** Digital and analogue meet through explicit conversions with well-defined event rules: sampling an analogue quantity into a digital value, holding a digital value as an analogue level, detecting a threshold crossing as a digital event, and quantizing. There are no implicit conversions between the two worlds.
- **One timeline.** Digital events and analogue integration share simulation time deterministically: a crossing that produces a digital event lands at a well-defined point in the delta-cycle order, the same every run.

Illustrative sketch: the syntax below is not designed yet.

```
entity Resistor<R: real> { p: Electrical inout, n: Electrical inout }

impl<R: real> Resistor<R> {
    let branch = p -> n;                    // the path from p to n
    branch.voltage == R * branch.current;   // holds at every instant
}

entity Capacitor<C: real> { p: Electrical inout, n: Electrical inout }

impl<C: real> Capacitor<C> {
    let branch = p -> n;
    branch.current == C * branch.voltage'ddt;
}
```

12.2 What it will not do

- **Change the digital language.** Digital semantics never depend on an analogue solver. A purely digital design compiles and simulates exactly as it does in Phase 1.
- **Share an IR.** Analogue models get their own intermediate representation and solver boundary. The digital IR stays exact and event-driven.
- **Accept analogue syntax early.** Until the phase lands, analogue constructs are errors in the digital compiler, so no design silently depends on something that does not work yet.

12.3 How it will be judged

Phase 2 is done when siox can solve small linear and nonlinear networks, co-simulate a digital controller with an analogue plant (a PWM controller driving a filter and a load, say), order bridge events deterministically, and report solver failures in terms of the source.

13 Phase 3: design and synthesis

Phase 3

The third phase turns component models into designs that existing tools can build. Components are what Phases 1 and 2 describe; **design** is connecting them into a product and realising it.

13.1 Projects and libraries

A real design is many files, libraries and vendors' IP. Phase 3 adds a project and package graph: reusable libraries with versions, discovered by name. The language names libraries and entities without saying whether their implementation is siox, VHDL, Verilog or a vendor database; that choice belongs to the project, not to the source. An instance of a vendor PLL and an instance of a siox counter are written the same way.

13.2 Foreign HDL, side by side

Most designs will mix siox with existing VHDL and Verilog: a vendor's IP core, a legacy block, a third-party interface. siox elaborates such a design with the foreign components as black boxes whose interface, metadata and library are known, links them by library metadata, and carries their names, widths, directions and clocks through to the output. Foreign HDL syntax never appears inside siox source, and compiling foreign HDL inside `sioxc` is only on the table if no existing compiled format can supply what is needed.

13.3 Constraints and physical intent

Clocks and their frequencies, I/O standards, pin assignments, placement and the base and vendor-namespaced attributes of § 11.3 are declared in the language, attached with `attr` bindings to the ports and instances they describe, checked against their declarations, and exported to each tool in its own format.

13.4 Synthesis through a vendor-neutral RTL

proposal

siox compiles to hardware before it compiles to another HDL (the compiler's `rtl-interface` proposal, summarized here). The design is lowered into a vendor-neutral, synthesis-facing RTL representation that describes elaborated hardware directly:

Object	What it records
modules, ports, instances	the elaborated hierarchy, with directions, widths and names
registers	clock, edge, reset kind and polarity, enable, next value
memories	width, depth and read/write ports, kept intact so tools infer block RAM
operations	`add`, `mul`, `mux`, comparisons with known widths and signedness, kept intact so tools map multipliers to DSP blocks
clock and reset domains	explicit, for CDC analysis, constraints and timing
metadata	vendor-neutral attributes such as `keep`, with vendor-specific ones namespaced
provenance	the source location and the construct (a macro, a pipeline) that produced each node

By the time a design reaches this representation, nothing of the source language is left to interpret: traits are resolved, generics specialised, macros expanded, attributes looked up, and pipelines turned into explicit registers. A backend only serialises:

- **SystemVerilog**, first, as a small, conservative subset that every tool accepts: modules, `always_ff`, `assign`, instances. The output reads like RTL assembly, not like hand-written code.
- **VHDL**, from the same representation, describing equivalent hardware.
- **CIRCT** and **Yosys**, where their own representations are a better fit.

A legalization step adapts the representation to each backend's subset; it changes form, never meaning, and a backend that cannot represent something faithfully says so rather than reinterpreting the design. `--emit rtl` shows the representation, answering the question every engineer asks of a high-level HDL: what hardware did it actually generate?

Simulation-only constructs (testbench `await`, file reads while running, random numbers, `after` delays) are reported when compiling for synthesis. Whether a construct may appear in hardware is answered per declaration by the target, not by excluding a library.

13.5 Schematics and boards

The last step is the board. Composing components into a netlist or schematic, with graphical tooling where it helps, connects the digital and analogue models of Phases 1 and 2 to the physical product, and keeps diagnostics, waveforms and debugging correlated with the source throughout.

13.6 How it will be judged

Phase 3 is done when a design containing siox and foreign components elaborates, emits a stable vendor-neutral artifact that preserves hierarchy, names, widths, directions, clocks and constraints, and that artifact goes through at least two synthesis ecosystems, with enough metadata coming back to correlate their reports with the source.

14 Tools, boundaries and status

14.1 The tools around the compiler

- **`sioxc`** is the compiler: one invocation per file, producing an object, a test executable or an inspection output. It is not a package manager or a test runner, and never will be.
- **A project tool** planned, in the role Cargo plays for Rust, will own projects, packages, dependencies and builds, calling `sioxc` underneath. It lives in its own repository.
- **`siox-lsp`** implemented, in its own repository, gives editors live diagnostics, navigation, completion and rename, using the compiler as a library.
- **GDB** implemented steps siox source in a test executable built with `-g`, with state named by its elaborated path.
- **The test corpus** implemented is a separate repository of runnable siox programs that every compiler change must pass, in both of the compiler's build configurations.
- **Syntax highlighting** implemented for siox code ships as a Sublime syntax definition (used by this document, Sublime Text and `bat`), coloured the way One Dark Pro colours Rust.

14.2 What siox will not be

- A package manager or test runner inside the compiler.
- A digital language whose meaning depends on an analogue solver.
- A home for vendor-specific primitives as language keywords.
- A container for foreign HDL syntax inside siox source.
- A language with conditional compilation that can hide code from the type checker; target selection is a value that both branches are checked against.
- A language that accepts constructs it cannot yet give meaning to.

14.3 How siox compares

	VHDL	SystemVerilog	Chisel	siox
Logic values	nine-value `std_logic`	four-value	two-value	nine-value IEEE, as library code
Widths	strict	implicit widening and truncation	inferred	strict; no implicit widening
Multiple drivers	resolved types	net resolution	last connection wins	only with a `Resolve` impl
Abstraction	generics, packages	classes, interfaces	Scala	traits, generics, views, operators as traits
Simulation	external event-driven simulators	external event-driven simulators	through generated Verilog	native code from the compiler
Testing	testbench entities	testbench modules and classes	Scala test frameworks	`#[test]` entities, test executables
Beyond digital	VHDL-AMS, a separate standard	—	—	analogue (Phase 2) and design (Phase 3) in one language

14.4 Where things stand

Area	Status
Modules, `use`/`type` with Rust's import forms, visibility	implemented

Area	Status
Entities, implementations, instances, instance arrays, views, generics	implemented
Structural generation with labelled scopes	implemented
Processes, wires, delta cycles, events, clocks, time	implemented
Nine-value logic, metavalues, numeric vectors, ranged integers, `natural`	implemented
Tuples, arrays with runtime indices and slices, run-time testbench arrays	implemented
Pattern matching: bit patterns, ranges, tuples, coverage proofs	implemented
Traits, operator traits, custom operators, literal affixes, type attributes	implemented
`From`, implicit `Into`, numeric promotion, `Default`	implemented
`#[derive(Eq, Ord)]`	implemented
Declared attributes, directives, lint levels	implemented
Testbenches, test executables, VCD/FST waveforms, GDB source debugging	implemented
`extern "C"`, compiler API, language server	implemented
User macros	implemented
`core`/`std` split, lang items, intrinsics	implemented
Fixed point; floating point with division, square root, fixed conversions	implemented
`Into` at port connections	partly implemented
Entity methods (associated functions; receiver methods paused)	partly implemented
Standard-library build-out (subnormals, rounding modes, linear algebra)	partly implemented
Compiler foundations (UI tests, JSON, one constant evaluator, resolved tree)	partly implemented
Pipelined functions	proposal
Implicit receivers, `static fn`, `static let`	partly implemented
Scoped blocks	proposal
Binary and unary custom operators	proposal
Parallel simulation	proposal
Co-simulation, VPI/VHPI and cocotb	proposal
Payload enums	on hold
Analogue and mixed signal	Phase 2
Projects, foreign HDL, constraints, vendor-neutral RTL, synthesis	Phase 3

14.5 Summary

siox is a bet that hardware deserves a language with VHDL's precision and Rust's structure, compiled by a toolchain as dependable as a modern software compiler, and that the same language can grow from digital logic to the whole circuit. The digital foundation, Phase 1, is complete and runs today. The analogue and design phases build on it without changing what the digital language means.

Appendix

15 Reference

15.1 Keywords

Items	`module` `use` `type` `pub` `extern` `entity` `impl` `struct` `view` `enum` `trait` `attr` `const` `let` `fn` `process` `macro`
Control	`if` `else` `match` `for` `in` `return`
Directions	`in` `out` `inout`
Contextual	`await` `after` `where` `self` `true` `false` `not` `_`
Word operators	`and` `or` `not`, and the standard library's `xor` `nand` `nor` `xnor`
Removed	`using` (now `use`/`type`), `wait` (now `await`); both are errors that name the replacement
Partial	Implicit receivers, `static fn`, `static let` (bare field names and entity methods to come) partly implemented

15.2 System attributes

`x'event`	`x` changed in this delta cycle
`x'old`	`x`'s value before the change
`x'length`	number of elements
`x'left`, `x'right`	the declared bounds, in written order
`x'high`, `x'low`	the larger and smaller bound
`x'ascending`	whether the declared range counts up
`x'range`	the declared range, for iteration: `for i in x'range`
`x'name`	a declared attribute: the bound value, or the default
`x'integers` …	a type attribute its implementation declares, such as the fixed formats' `'integers` and `'fractions`

`'ddt` is reserved for Phase 2 and is an error today.

15.3 Directives

`#[test]`	compile this entity into the test executable
`#[allow(lints)]`	do not report these lints in the next item or statement
`#[warn(lints)]`	report them as warnings (the default)
`#[deny(lints)]`	report them as errors
`#[forbid(lints)]`	as `deny`, and nothing inside may lower it
`#![level(lints)]`	any lint level, for the whole module
`#[inline(always)]`	expand this function's calls in the simulation
`#[inline(never)]`	call one shared copy of this function in the simulation
`#[inline]`	a hint to expand
`#[derive(Eq, Ord)]`	write a struct's comparison impls, field by field
`#[latched(N)]`	a pipelined function of depth `N` proposal
`#[latch]`	one pipeline stage: a block, or the statement that follows proposal

15.4 Lints

Lint	Code	Reports
`possible_latch`	W-P002	a signal assigned on some paths of a combinational process only
`unused_signal`	W-P003	a signal nothing reads
`unused_param`	W-P004	a parameter nothing uses
`unused_import`	W-P005	an import nothing uses
`unreachable_match_arm`	W-P006	a `match` arm earlier arms already cover
`non_exhaustive_match`	W-P007	a `match` that misses values
`suspicious_logic_compare`	W-P008	a comparison of logic values that is likely a mistake
`suspicious_reset`	W-P009	a reset pattern that is likely a mistake
`combinational_loop`	W-P010	a signal that feeds itself with no register in the path
`undriven_output`	W-P011	an output or signal nothing drives
`unconnected_input`	W-P012	an instance input left unconnected
`dead_assignment`	W-P014	an assignment a later one always overrides
`unimplemented_attr`	W-P015	an attribute no tool reads yet
`incomplete_struct_literal`	W-P016	a struct literal that leaves fields at their defaults
`unknown_lints`	W-P017	a lint name that does not exist
`overlapping_range_endpoints`	W-P018	two `match` ranges sharing exactly one value, usually an off-by-one
`non_constant_loop_bound`	W-P019	a loop that simulates but cannot unroll into hardware
`renamed_default`	W-P020	`T::new()` on a type with no `new`: write `T::default()`
`unwired_instance`	W-P021	an instance-array slot nothing connects
`warnings`	—	every lint at once

15.5 Glossary

Delta cycle One step of simulation within a single instant: every process whose inputs changed runs, their writes are applied together, and the step repeats until nothing changes.

Driver A source of a signal's value: a wire, a process that assigns it, or an instance output connected to it.

Elaboration Turning parameterized entities into a concrete tree of instances, with generation unrolled and parameters substituted.

Entity A hardware component's interface: its ports.

Metavalue A logic value other than plain `'0'`/`'1'`, such as `'U'`, `'X'` or `'Z'`.

Process A block of ordered statements that runs concurrently with everything else.

Promotion Two numbers of different types meeting, in arithmetic, at the higher rung of one ladder in a format that holds both exactly.

Resolution Folding the values of several drivers of one signal into one, defined by the signal type's `Resolve` implementation.

Tuple An anonymous struct whose fields are its positions: `(A, B)`, read `t.0`, `t.1`.

View A directional role over a struct, giving each field a direction, so one struct serves both ends of a bus.

15.6 Further reading

- The reference documents (the language specification, the standard-library reference, simulation, testing, architecture and the Phase 1 audit), kept beside this paper: https://github.com/Siox-lang/siox-paper/tree/main/docs
- The siox compiler and its proposals: https://github.com/Siox-lang/sioxc

- The siox test corpus: https://github.com/Siox-lang/siox-tests
- IEEE 1076-2019, the VHDL standard, for the nine-value logic system.
- Spade, an expression-based HDL with pipelines: https://spade-lang.org/
- The Rust Reference and the rustc development guide, for the conventions siox follows in its structure and its compiler.